APURV BHUSHAN

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EDUCATION

Btech - Robotics and Autonomous Systems	Plaksha University, Mohali	CGPA- 8.56	2026
12th - CBSE	Lotus Valley International School, Gurgaon	Percentage- 93.4	2024
10th - CBSE	Lotus Valley International School, Gurgaon	Percentage- 93.6	2022

EXPERIENCE

PocketQube Satellite | CanSat Competition 2026-27

- As part of the Mechanical Subsystems team, was tasked with designing Solar Panel & Parachute release mechanism along with the design of the PocketQube satellite as part of CANSAT's 2026-27 mission statement.
- Tested prototypes for the solar panel release mechanism.

Airfoil Aerodynamics | Prof. Andy Ruina May 2026- June 2026

- Extracted the datasets from a 1981 paper by Sheldahl and Klimas on lift, drag and moment coefficients of the following airfoil profiles- NACA-0009, -0012, -0012H, 0015.
- Their work was on performing wind tunnel experiments on the listed airfoil profiles and generating data for a thicker set of airfoil profiles (-0018, -0021, -0025) using a software called PROFILE.
- Digitized all lift vs alpha, drag vs alpha, moment vs alpha, axial force vs alpha graphs from the paper using the extracted datasets.
- Collated all digitized information in a folder for Professor Andy Ruina to use for his research on flat plate dynamics.
- Wrote a simplified version on latex of 1981 paper containing all newly digitized plots.

Paraglider Release and Control System | CanSat Competition 2025-26 Nov 2025- Feb 2026

- Helped design the paraglider release system and paraglider descent control system as a part of the Mechanical Design Team in Kalki (Unviversity's Team for CanSat 2026).
- The paraglider used was a modified low-aspect ratio ram-air parafoil providing greater stability for flight
- The descent control system made use of DC micromotors and pulleys to control direction of the payload descent.

Portable Makerspace | RedBrick Hacks Ashoka Dec 2025- Jan 2026

- Designed attachments for power tools in order for these tools to function as other machines, such as a Drill-Lathe, Jigsaw-Scrollsaw, Circular saw-Tablesaw and designed a CAD model for a custom workbench as a part of a 4-member team working on designing a low-cost portable makerspace. Awarded 3rd place.

Intern | Plaksha Makerspace June 2025- July 2025

- Developed a cyclorotor prototype leveraging makerspace resources including 3D printers and laser cutting machines, strengthening hands-on CAD and mechanical design skills.
- Collaborated with a fellow intern to build a power supply from scratch, gaining practical exposure to electronic components such as transformers, buck converters, and rectifiers.

PROJECTS

AdaGamma Optimizer | Course Project | Prof. Ashish Kumar Pandey Oct 2025- Nov 2025

- Derived the mathematical formulation for AdaGamma, extending the Adam optimizer by introducing skewness and variance terms to improve performance on skewed and non-Gaussian data distributions, as part of a 4-member team.
- Achieved 50–70% faster convergence and higher final accuracy compared to Adam, demonstrating strong generalization and robustness across diverse loss landscapes.

Cyclorotor Autogyro Project | Prof. Shashikant Pawar, Prof. Andy Ruina Sep 2025- Present

- Project initially started as a recreation of the 4-bladed cyclorotor for underwater applications but has transformed into exploring an autogyro concept. Read 20+ papers to conduct a literature review to understand the science behind the cyclorotor and to identify potential gaps to be filled in this unorthodox domain of propulsion systems.
- **Stage 1:** Conducted Flat plate 2D CFD simulations with the help of FreeCAD and OpenFOAM at various angles of attack to understand the simulation process and used the quantitative data to benchmark against existing data.
- **Stage 2:** Went through an intensive design iteration process for designing a 3-axis Wind Tunnel Force Balance to measure aerodynamic forces and moments on a static and autorotating flat plate.
- **Further Stages** include assembling, calibrating and conducting wind tunnel experiments on a flat plate and a light-weight fixed pitch cyclorotor

COURSEWORK

Dynamics and Simulation | Elective Course Offered by Prof. Andy Ruina Aug 2026- Dec 2026

- Master's level course on solving differential equations of dynamical systems numerically.
- 3 methods to solve DEs- Newton-Euler, Differential Algebraic Equations, Legrangian Method.
- Covered concepts like Simple Pendulum numerics (cartesian and polar), 2D Dumbell, Double Pendulum, Triple Pendulum, Rotation with zero angular momentum, Inverted pendulum with shaking base, Constrained Motion, Rotation Matrices, etc.

Robot Motion Simulation and Optimization | Elective Course offered by Prof. Andy Ruina Jan 2026- May 2026

- Course on solving differential equations numerically- more focused on optimization.
- Introduced to ODE45, Euler method
- Covered concepts like 2D & 3D Spring Mass Problems, Canon Ball with Air drag problem, Stability of Systems, Periodic motion (Poincare Section, stability of periodic motions), Initial Value Problem using collocation, Boundary Value Problem, Collocation- Euler and Trapezoidal Discretization, Optimal Planning and Control, MPC Tracking, Time-Optimal Control.

Data Observer and Design | Elective Course Offered by Prof. Ravi Banavar Jan 2026- May 2026

- Studied and applied state estimation techniques, including sensor fusion, recursive estimation, Kalman gain optimization, and covariance propagation.
- Implemented and tuned Kalman Filters for IMU-based attitude estimation, fusing accelerometer and gyroscope measurements; investigated the effects of process and measurement noise on estimation accuracy and stability.
- Implemented and compared nonlinear attitude estimation algorithms on SO(3), including gyro integration, TRIAD, and Mahony complementary filters, using accelerometer, gyroscope, and magnetometer data for 3D orientation tracking and online gyro bias correction.

System Dynamics and Control | RAS Core | Prof. Sunita Chauhan Jan 2026- May 2026

- Course on Control Theory and System Modelling
- Covered concepts like Laplace Transform, Fourier Series, Mass Spring Damper Systems, Performance Analysis & Controller Design- Root Locus diagram, Bode Plots (Up to Second Order).
- Course Mini Project- Follower-Leader System. Tasked with making an autonomous bot that would make use of PID control to follow a remote controlled 'Leader' Robot

SKILLS

- 3D Modelling/ 3D printing- Proficiency in **Onshape, FreeCAD**. Working knowledgeof **Fusion360**
- Programming & Simulation - **MATLAB** and **Python.**
- CFD Simulation- **OpenFOAM** (FreeCAD's cfdof Workbench)
- Technologies- **Machine Learning**
- Laser Cutting- Working knowledge of **RDWorks.**
- Woodworking- Worked with machines like **Bandsaw, Lathe, Table saw, Electric Hand Planer**

ACHIEVEMENTS/EXTRACURRICULARS

- Secured **98.6 percentile** in JEE Mains 2024.
- Part of the **Plaksha Cricket Team** as an opening batsman and wicket-keeper. Participated in **UPES' Ojas 2025 Tournament** and **Panjab University's Umang 2026 Tournament.**
- **RC Plane Hobbyist**. Built and Flew aircraft alongside other hobbyists.

POSITIONS OF RESPONSIBILITY

Captain, Plaksha Cricket Team Sep 2026- Present

Club Kalki (Robotics Club) Member | Plaksha University: Oct 2025- Present

- Part of the Mechanical Subsystems Team for the CANSAT Competition two years in a row (2025-26 and 2026-27 edition of the CANSAT Compeition)

Makerforce Core Member | Plaksha University: Dec 2024- Present

- Served as a point of contact for 2 teams participating in the intra-university, Rube Goldberg Competition 2025, a 24-hour event and coordinated execution efforts such as setting up a gaming zone where participants would take turns to play on the WII console as a destressor.
- Makerspace guide for 20+ external counsellors, teachers and students.